

Architecture Selection — Layered Architecture

Patient Health Record Consent Management System

The selected architecture for the Patient Health Record Consent Management System is **Layered Architecture**. The system is divided into Presentation, Business Logic, and Data layers, allowing each layer to focus on a specific responsibility.

The first reason for selecting Layered Architecture is **clear separation of responsibilities**. The Presentation Layer handles interactions with patients and clinic administrators. The Business Logic Layer manages consent creation, validation, and revocation. The Data Layer handles health-record access and audit information. This separation makes the system easier to understand and maintain.

The second reason is **maintainability**. Changes to the patient or clinic interface can be made without directly modifying the consent-management logic. Similarly, data-related changes can be isolated within the Data Layer. This structured organization is appropriate for a patient-centric healthcare system.

From a **security** perspective, the Business Logic Layer can enforce authorization and consent verification before a health record is provided. A clinic administrator therefore cannot access patient records unless a valid, non-expired consent permission exists. Consent grant, revoke, and access events can also be recorded by the Audit Trail Component.

From a **performance** perspective, the layered structure provides a predictable processing path from the user interface through business logic to the required data operation. Consent verification can be performed before record retrieval, avoiding unauthorized processing and supporting efficient request handling.

Therefore, Layered Architecture is selected because it provides clear separation of concerns, maintainability, controlled authorization, and a straightforward structure suitable for the Patient Health Record Consent Management System.